

The Second Connectivity Tax

Collision entropy, Bethe total correlation, and a universal 0.462 lower bound

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Abstract

For a finite nonempty tree T , let $\mu(T)$ be the mean size of a uniformly random independent set and let $\alpha(T)$ be its independence number. Write

$$a_*(1) = \inf_T \frac{\mu(T)}{\alpha(T)}.$$

Starting from the exact entropy-partition decomposition, we add a second connectivity tax. The first tax is the collision term already visible under the comparison hard-core measure. The second is the total correlation of the conditional factor model on the isolated B -vertices. Because this conditional factor graph is a forest, its entropy is exactly Bethe and the total correlation decomposes as a sum of factor multi-informations. A local discharging argument then shows that the only potentially dangerous scalar degrees are 3 and 4, and that the critical clean factor is the pair $(4, 4)$. The remaining contamination is absorbed by the B -core edge budget. We obtain the exact universal bound

$$a_*(1) \geq \frac{231}{500} = 0.462.$$

All nontrivial scalar inequalities in the proof reduce to rational interval arithmetic for logarithms and are checked by the accompanying standard-library verifier `verify_second_connectivity_0462.py`; no floating-point operation is used in the certificate.

1 Statement

For a finite graph G , put

$$Z_G = |\mathcal{I}(G)|, \quad \mu(G) = \frac{1}{Z_G} \sum_{I \in \mathcal{I}(G)} |I|, \quad \alpha(G) = \max_{I \in \mathcal{I}(G)} |I|.$$

Let

$$c := \frac{\log(34/5)}{6 \log 2}, \quad \gamma := \frac{1}{2} - c, \quad r_0 := \frac{231}{500}.$$

Thus

$$\gamma - (r_0 - c) = \frac{1}{2} - r_0 = \frac{19}{500}.$$

Theorem 1 (Second connectivity tax). *Every finite nonempty tree T satisfies*

$$\mu(T) \geq \frac{231}{500} \alpha(T).$$

Consequently

$$a_*(1) \geq 0.462.$$

The proof is uniform in the order and maximum degree of T .

2 The adjusted entropy decomposition

Fix a maximum independent set A of T and write

$$\alpha = |A|, \quad B = V(T) \setminus A, \quad K = T[B], \quad e = |E(K)|.$$

For $v \in B$ put

$$d_v = |N(v) \cap A| \geq 1.$$

Since T is a tree,

$$\sum_{v \in B} (d_v - 1) = \alpha - 1 - e. \quad (1)$$

For $d \geq 1$ define

$$\ell_d := \left[\log \frac{5/4}{1 + 2^{-d}} \right]_+, \quad \Delta_d := \gamma(d - 1) - \frac{\ell_d}{2 \log 2}.$$

The entropy-partition decomposition gives

$$\mu(T) - c\alpha \geq \gamma(1 + e) + \sum_{v \in B} \Delta_{d_v} + \frac{R_A}{2 \log 2} + \frac{D_{\text{KL}}(P \| Q_{1/4})}{2 \log 2}, \quad (2)$$

where P is the law of $J = I \cap B$ under the uniform independent-set measure, $Q_{1/4}$ is the hard-core measure of activity $1/4$ on K , and

$$R_A = \log \mathbb{E}_Q 2^{O(J)}, \quad O(J) = \sum_{a \in A} (|J \cap N(a)| - 1)_+.$$

We shall not use the additional nonnegative coordinate-comparison remainder.

Subtract $(r_0 - c)\alpha$ and use (1). Define the adjusted scalar charge

$$s_d := \Delta_d - (r_0 - c)(d - 1) = \frac{19}{500}(d - 1) - \frac{\ell_d}{2 \log 2}. \quad (3)$$

Then

$$\mu(T) - r_0\alpha \geq \frac{19}{500}(1 + e) + \sum_{v \in B} s_{d_v} + \frac{R_A}{2 \log 2} + \frac{D_{\text{KL}}(P \| Q_{1/4})}{2 \log 2}. \quad (4)$$

Write

$$D_d := (-s_d)_+, \quad t_d := \frac{(s_d)_+}{d}.$$

Thus

$$s_d = d t_d - D_d. \quad (5)$$

Lemma 2 (Only degrees 3 and 4 are dangerous). *One has*

$$D_d = 0 \quad (d \neq 3, 4),$$

and

$$0 < D_3 < D_4.$$

Moreover, for every $d \geq 6$,

$$t_d \geq D_4.$$

Proof. This is a one-dimensional calculation from (3). The finite cases reduce to rational logarithmic inequalities. For the tail, use $\ell_d < \log(5/4)$ and the linear growth of $(19/500)(d - 1)$. The exact interval audit verifies all cases through $d = 80$ and the endpoint of the monotone tail. $\square$

Numerically, only for orientation,

$$D_4 = 0.00323262681851147 \dots \quad (6)$$

No numerical value is used in the proof.

3 The collision term on isolated B -vertices

Let

$$R := \{v \in B : \deg_K(v) > 0\}, \quad C := B \setminus R.$$

Every vertex of C is isolated in K . Under the comparison measure Q with activities $w_v = 2^{-d_v}$, the indicators of vertices in C are therefore mutually independent and

$$\Pr_Q(v \in J) = p_{d_v} := \frac{1}{2^{d_v} + 1}.$$

For $a \in A$, let

$$X_a = |J \cap N(a) \cap C|.$$

Since

$$O(J) \geq \sum_{a \in A} (X_a - 1)_+,$$

and the functions $2^{(X_a - 1)_+}$ are increasing functions of independent Bernoulli variables on C , Harris' inequality gives

$$\frac{R_A}{2 \log 2} \geq \sum_{a \in A} \mathcal{R}_a, \tag{7}$$

where, if the C -neighbour degrees at a are $d_1, \dots, d_q$,

$$\mathcal{R}(d_1, \dots, d_q) := \frac{1}{2 \log 2} \log \frac{\prod_{i=1}^q (1 + p_{d_i}) + \prod_{i=1}^q (1 - p_{d_i})}{2}. \tag{8}$$

Indeed,

$$\mathbb{E} 2^{(X-1)_+} = \frac{1}{2} \left(\prod_i (1 + p_i) + \prod_i (1 - p_i) \right).$$

For the critical pair $d_1 = d_2 = 4$, set

$$\kappa := \mathcal{R}(4, 4) = \frac{\log(290/289)}{2 \log 2}. \tag{9}$$

4 Conditional Bethe total correlation

Condition on the configuration on R . Under $Q_{1/4}$, the variables in C remain independent Bernoulli(1/5) variables and are independent of R . Under P , the conditional law on C factorizes on the bipartite forest with variable nodes C and factor nodes A . At a factor a , if some conditioned neighbour in R is occupied the factor is constant; otherwise the active factor is

$$\psi_a(x_{N_C(a)}) = \begin{cases} 2, & x_v = 0 \text{ for every } v \in N_C(a), \\ 1, & \text{otherwise.} \end{cases}$$

The factor graph is a forest because it is a subgraph of T .

For any probability distribution factorizing on a factor forest, the Bethe entropy identity is exact. Hence, conditionally on R ,

$$\text{TC}(P_{C|R}) = \sum_{a \text{ active}} \left(\sum_{v \in N_C(a)} H(S_v) - H(S_{N_C(a)}) \right). \tag{10}$$

Since the reference measure on C is a product,

$$D_{\text{KL}}(P \| Q_{1/4}) \geq \mathbb{E}_R \text{TC}(P_{C|R}). \tag{11}$$

We only need a quantitative bound for an active factor with exactly two dangerous neighbours. After all other factors are absorbed into cavity messages, its two-variable law has unnormalised weights

$$(2, a, b, ab), \quad \frac{1}{8} \leq a, b \leq 1,$$

because a dangerous vertex has degree 3 or 4, and every other incident active factor changes its cavity odds by a factor between $1/2$ and 1 .

Lemma 3 (Critical pair mutual information). *For the law with weights $(2, a, b, ab)$ on $\{0, 1\}^2$ and $1/8 \leq a, b \leq 1$, the mutual information is minimized at $a = b = 1/8$. Therefore*

$$\frac{I(X; Y)}{2 \log 2} \geq \iota, \tag{12}$$

$$\iota := \frac{1}{2 \log 2} \frac{1}{145} \left[128 \log \frac{290}{289} + 16 \log \frac{145}{153} + \log \frac{145}{81} \right]. \tag{13}$$

Numerically $\iota = 0.000821638641763987 \dots$

Proof. The joint probabilities at $a = b = 1/8$ are

$$\frac{1}{145}(128, 8, 8, 1),$$

with marginals $(136, 9)/145$, which gives (13). It remains to locate the minimum. If $I(a, b)$ denotes the mutual information, direct differentiation gives

$$\frac{\partial I}{\partial a} = -\frac{N(a, b)}{(ab + a + b + 2)^2},$$

up to a further positive factor, where

$$N(a, b) = (b + 1)(b + 2) \log \frac{b + 1}{b + 2} + b \log \frac{a + 1}{a + 2} + 2(b + 1) \log 2.$$

For fixed b , $N(a, b)$ is increasing in a , so it is enough to take $a = 1$. Put $F(b) = N(1, b)$. Then

$$F'(b) = 1 + \log \left[\frac{8}{3} \left(\frac{b + 1}{b + 2} \right)^{2b+3} \right].$$

The bracket is increasing on $[1/8, 1]$ and at $b = 1$ equals $256/729 < e^{-1}$; hence $F'(b) < 0$. Thus $F(b) \leq F(1/8) < 0$, the last strict endpoint comparison being one of the exact logarithmic checks in the audit. Therefore $\partial I / \partial a \geq 0$. By symmetry the same holds for b , and the minimum is attained at $(1/8, 1/8)$. $\square$

For later use define

$$\zeta := 2D_4 - \kappa - \frac{\iota}{2}. \tag{14}$$

The audit proves $\zeta > 0$; numerically

$$\zeta = 0.00356273055901312 \dots$$

5 Local discharging

A factor is called *clean* if it has no neighbour in R , and *contaminated* otherwise. The positive scalar charge of a vertex $v \in C$ is distributed equally over its d_v incident A -factors, contributing t_{d_v} to each incidence. By (5), the only unpaid vertex charges are the deficits D_3, D_4 .

Root every connected component of the clean factor forest at a C -vertex. At a clean factor, one incident C -vertex is the parent and all remaining incident C -vertices are children.

Lemma 4 (Clean-factor inequality). *For every clean factor a ,*

$$\mathcal{R}_a + \mathcal{I}_a + \sum_{v \in N_C(a)} t_{d_v} \geq \sum_{v \in \text{children}(a)} D_{d_v}, \quad (15)$$

where $\mathcal{I}_a$ is the factor multi-information from (10). It is enough to use $\mathcal{I}_a = 0$ except when a has exactly two neighbours and both have degree 3 or 4, in which case $\mathcal{I}_a/(2 \log 2) \geq \iota$.

Proof. By Lemma 2, a neighbour of degree at least 6 already contributes $t_d \geq D_4$ and can be stripped off together with at most one dangerous child. After this reduction, only degrees $1, \dots, 5$ remain. For arity at least 7, the marginal increase of the collision term from an additional dangerous neighbour already exceeds D_4 , so the residual problem reduces to arities at most 6. The remaining finite family is checked exactly by rational logarithmic intervals. The unique critical configuration is the pair $(4, 4)$, where

$$\kappa + \iota - D_4 > 0.$$

The exact audit gives a strict margin exceeding 8.07×10^{-5} . □

Thus every clean component pays all dangerous deficits except possibly that of its root, which is at most D_4 .

Now consider a contaminated factor. Let $k_R \geq 1$ be its number of neighbours in R . Its C -neighbours lie in distinct clean components whenever those components are attached through this factor, because the original graph is a tree.

Lemma 5 (Contaminated-factor inequality). *After paying the distributed positive credits t_d and the collision contribution $\mathcal{R}_a$, the total unpaid dangerous-root deficit attached through a contaminated factor is at most*

$$\zeta k_R.$$

If $k_R = 1$ and the unique critical C -configuration is the pair $(4, 4)$, one additionally uses half of the pair mutual-information bound ι .

Proof. If $k_R = 1$, the unique R -neighbour is unoccupied with probability at least $1/2$: deleting an occupied vertex injects independent sets containing it into independent sets avoiding it. Hence in the critical $(4, 4)$ case the factor is active with probability at least $1/2$, giving the additional contribution $\iota/2$. The worst residual is then exactly

$$2D_4 - \kappa - \frac{\iota}{2} = \zeta.$$

All other residual configurations are strictly smaller. For $k_R \geq 2$ the right side only increases, and no total-correlation contribution is needed. As in Lemma 4, degrees at least 6 and arities at least 7 reduce monotonically to a finite audit over degrees $1, \dots, 5$ and arities at most 6. □

6 The R -core budget

Let $k_v = \deg_K(v)$ for $v \in R$. Then $k_v \geq 1$ and

$$\sum_{v \in R} k_v = 2e.$$

Every contaminated factor is incident with at least one R -vertex. Summing Lemma 5 therefore leaves total unpaid clean-component root deficit at most

$$\zeta \sum_{v \in R} d_v. \tag{16}$$

The R -vertices themselves have not spent their scalar charge. The edge term in (4) distributes as

$$\frac{19}{500}e = \sum_{v \in R} \frac{19}{1000}k_v.$$

The following one-variable inequality closes the argument.

Lemma 6 (R -core domination). *For every integer $d \geq 1$ and every integer $k \geq 1$,*

$$s_d + \frac{19}{1000}k \geq \zeta d. \tag{17}$$

Proof. The left side is increasing in k , so it suffices to take $k = 1$. The exact audit checks $1 \leq d \leq 80$. For the tail, $s_d/d \rightarrow 19/500$ while $\zeta < 19/500$, and the endpoint estimate is already strict. The smallest finite margin occurs at $d = 4$ and is positive. $\square$

Summing (17) over R gives

$$\sum_{v \in R} s_{d_v} + \frac{19}{500}e \geq \zeta \sum_{v \in R} d_v, \tag{18}$$

which pays (16) in full.

7 Proof of Theorem 1

Apply (4). On C , decompose every scalar charge as $d_v t_{d_v} - D_{d_v}$ and distribute the positive credits over the incident A -factors.

For each clean component, Lemma 4 pays every dangerous deficit except possibly one root deficit. If $R = \emptyset$, the whole tree has only one such component, and the unused constant term $19/500$ is much larger than D_4 , so the root is paid.

If $R \neq \emptyset$, every clean component is attached to the rest of the tree through at least one contaminated factor. Root the quotient forest toward the R -core. Lemma 5 pays all child-component root deficits up to the residual (16). Lemma 6, summed as in (18), pays that residual and simultaneously pays all scalar charges on R .

All terms retained from (4) have now been exhausted without a negative remainder. Therefore

$$\mu(T) - \frac{231}{500}\alpha(T) \geq 0,$$

which proves the theorem. $\square$

8 Exact audit and scope

The script `verify_second_connectivity_0462.py` uses only `fractions.Fraction`. Logarithms are enclosed by the exact atanh series

$$\log x = 2 \sum_{j=0}^{N-1} \frac{y^{2j+1}}{2j+1} + R_N, \quad y = \frac{x-1}{x+1},$$

with the rational remainder bound

$$|R_N| \leq \frac{2|y|^{2N+1}}{(2N+1)(1-y^2)}.$$

It verifies:

- degrees 3 and 4 are the only negative adjusted scalar charges;
- $t_d \geq D_4$ for every $d \geq 6$;
- the complete reduced clean-factor table;
- the complete reduced contaminated-factor table;
- the exact critical identity defining ζ ;
- the R -core domination inequality and its tail.

The executed audit terminates with **PASS**. It is a finite exact certificate for the scalar reductions used above; it is not a proof-assistant formalization.

Remark 7 (Numerical location of the next barrier). *The critical clean pair $(4, 4)$ has local budget*

$$\kappa + \iota = 0.00331334239889181 \dots,$$

whereas at $r_0 = 0.462$ its required deficit is

$$D_4 = 0.00323262681851147 \dots$$

Thus this second-tax mechanism still has positive local room. If all other stability losses could be eliminated, the clean $(4, 4)$ pair alone would permit the formal value

$$c + \frac{\kappa + \iota}{3} = 0.46202690519346 \dots$$

We do not claim this latter number as a universal bound here.